Accessible Technology Handbook

A short guide to assistive tools for people with visual impairments

Introduction

Digital accessibility means designing documents, websites and software so that everyone can use them, including people who are blind or have low vision. Screen readers, magnifiers, braille displays and text-to-speech tools turn visual content into sound, enlarged text or touch.

This handbook is a test document for an offline document reader. It contains headings, paragraphs, lists, tables, captions and repeated headers and footers, so that every part of a reading pipeline can be checked.

Why accessibility matters

According to the World Health Organization, at least 2.2 billion people worldwide have a near or distance vision impairment. Many of them rely on assistive technology every day to read mail, study, work and move around safely.

Chapter 1: Screen Readers

A screen reader is software that reads out loud what appears on the screen. It follows the structure of a document, so headings, lists and tables must be marked up correctly for it to work well.

Common screen readers

- NVDA is a free and open-source screen reader for Windows.
- JAWS is a commercial screen reader for Windows.
- VoiceOver is built into macOS and iOS devices.
- TalkBack is built into Android devices.

Navigation shortcuts

Users move through a document by headings, links and landmarks rather than line by line. Pressing the H key in NVDA jumps to the next heading, which can save several minutes in a long report.

> *“A document that cannot be navigated by structure is, for a blind reader, a wall of sound.”*

Chapter 2: Optical Character Recognition

Optical character recognition, or OCR, converts images of text into machine-readable text. It is essential for scanned books and photographed letters, which contain no real text for a screen reader to speak.

How OCR works

1 The page image is converted to grayscale.

2 The image is straightened and thresholded to separate text from background.

3 Lines and words are detected and segmented.

4 Each character is recognised and the text is assembled in reading order.

Comparison of common tools

Tool	Type	Cost	Works offline
NVDA	Screen reader	Free	Yes
JAWS	Screen reader	Paid	Yes
Tesseract	OCR engine	Free	Yes
Cloud OCR service	OCR engine	Paid	No

Table 1: Assistive tools compared by type, cost and offline support.

Tesseract was first developed in the 1980s and is maintained as an open-source project. Accuracy depends strongly on scan quality: a clean 300 DPI scan typically gives far better results than a blurry photo.

Chapter 3: Text-to-Speech

Text-to-speech, often shortened to TTS, converts written text into spoken audio. Modern offline engines run entirely on the user’s own computer, so no internet connection or account is needed.

Choosing a voice and speed

Experienced screen reader users often listen at speeds above 300 words per minute, while beginners usually prefer about 150 words per minute. A good reader should let the user change the speed at any moment.

Reading controls

- Pause and resume reading with a single key.
- Jump to the next or previous heading.
- Skip headers, footers and page numbers automatically.
- Announce the page number when a new page begins.

Privacy

Offline processing protects privacy. Medical letters, bank statements and personal mail never leave the device, which is a major advantage over cloud-based services.

Chapter 4: Design Guidelines

The Web Content Accessibility Guidelines, known as WCAG, describe how to make content accessible. Version 2.2 organises the requirements under four principles: perceivable, operable, understandable and robust.

Key recommendations

1 Provide text alternatives for every image.

2 Ensure a colour contrast ratio of at least 4.5 to 1 for normal text.

3 Make all functions available from the keyboard.

4 Use real headings and lists instead of visual formatting alone.

5 Test with a real screen reader and with real users.

Conclusion

Accessible design benefits everyone. A reader that works offline, speaks clearly, respects document structure and answers simple questions about the text can give blind and low-vision users real independence.

In summary, good accessibility combines correct document structure, reliable OCR, natural speech output and careful testing with the people who will use the tools.